

LESSON 5-3 • SESSION 2 • PREREQUISITE PRACTICE B • SKILL 1 OF 2

Decompose Trapezoids

Builds on Lesson 5-3 Session 1 • 6.G.A.1

TODAY'S LESSON

The formula gives the same area as cutting a trapezoid into a rectangle and two triangles.

YOU NEED TO

Find a trapezoid's area by cutting it into a rectangle and triangles. **You will use it in Practice Form A, problems 3 and 7.**

STEPS TO FOLLOW*use these same steps on every problem*

- 1** Find the **rectangle**: short base $\times$ height.
- 2** Find the **triangles**: $(\text{long} - \text{short}) \times \text{height} \div 2$.
- 3** **Add** the two areas.

WATCH ME*the steps in action*

A trapezoid has bases 10 ft and 18 ft and height 6 ft. Find its area.

Step 1 $10 \times 6 = 60$.

Step 2 $(18 - 10) \times 6 \div 2 = 24$.

Step 3 $60 + 24 = 84$.

Answer **84 ft²**

TRY IT WITH HELP*fill in each step*

- 1.** A trapezoid has bases 8 m and 12 m and height 5 m. Find its area.

Step 1 $8 \times 5 = \underline{\hspace{2cm}}$

Step 2 $(12 - 8) \times 5 \div 2 = \underline{\hspace{2cm}}$

Step 3 $40 + 10 = \underline{\hspace{2cm}}$ m²

- 2.** A trapezoid has bases 6 in and 14 in and height 4 in. Find its area.

Step 1 $6 \times 4 = \underline{\hspace{2cm}}$

Step 2 $(14 - 6) \times 4 \div 2 = \underline{\hspace{2cm}}$

Step 3 $24 + 16 = \underline{\hspace{2cm}}$ in²

YOUR TURN*use the steps on your own*

- 3.** Cut each trapezoid apart. Find its area.

a. bases 4 and 10, $h = 3$ _____

b. bases 5 and 9, $h = 6$ _____

c. bases 10 and 16, $h = 4$ _____

d. bases 7 and 15, $h = 2$ _____

Find a Missing Dimension

Builds on Lesson 5-2 Session 2 • 6.G.A.1

TODAY'S LESSON

A trapezoid has an area of 84 m^2 and a height of 7 m. You work backward to find a base.

YOU NEED TO

Work backward from an area: double it, divide by the height, then subtract.
You will use it in Practice Form A, problem 8.

STEPS TO FOLLOW*use these same steps on every problem*

- 1 Double** the area.
- 2 Divide** by the height. That is the sum of the bases.
- 3 Subtract** the base you know.

WATCH ME*the steps in action*

A trapezoid has area 60 cm^2 and height 5 cm. One base is 4 cm. Find the other base.

Step 1 $60 \times 2 = 120.$

Step 2 $120 \div 5 = 24.$

Step 3 $24 - 4 = 20.$

Answer **20 cm**

TRY IT WITH HELP*fill in each step*

- 4.** A trapezoid has area 40 m^2 and height 4 m. One base is 7 m. Find the other base.

Step 1 $40 \times 2 = \underline{\hspace{2cm}}$

Step 2 $80 \div 4 = \underline{\hspace{2cm}}$

Step 3 $20 - 7 = \underline{\hspace{2cm}} \text{ m}$

- 5.** A trapezoid has area 90 ft^2 and height 10 ft. One base is 8 ft. Find the other base.

Step 1 $90 \times 2 = \underline{\hspace{2cm}}$

Step 2 $180 \div 10 = \underline{\hspace{2cm}}$

Step 3 $18 - 8 = \underline{\hspace{2cm}} \text{ ft}$

YOUR TURN*use the steps on your own*

- 6.** Find the other base.

a. $A = 36$, $h = 4$, one base 5 $\underline{\hspace{2cm}}$

b. $A = 50$, $h = 5$, one base 6 $\underline{\hspace{2cm}}$

c. $A = 70$, $h = 7$, one base 9 $\underline{\hspace{2cm}}$

d. $A = 24$, $h = 3$, one base 10 $\underline{\hspace{2cm}}$

READY CHECK*a problem like the one in today's lesson*

- 7.** A trapezoid has an area of 48 m^2 and a height of 6 m. What is the sum of its bases?

A 8 m

B 16 m

C 42 m

D 288 m

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about # $\underline{\hspace{1cm}}$